Ayush Kuriakose Chacko

Bengaluru, Karnataka, India

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Summary

AI/ML Engineer with production experience at Mercedes-Benz R&D, building **predictive modeling, computer vision, and generative AI** systems from prototype to deployment. Focused on making large models fast and cheap enough for real-world constraints, from **edge inference to LLM cost optimization**, with a track record of measurable, benchmarked improvements. Proficient in Python and PyTorch.

Experience

Mercedes-Benz Research and Development India **December 2025 - September 2026**
Graduate Apprentice Trainee *Bengaluru, Karnataka*

- Developed a real-time **gesture recognition system** for autonomous parking, training an **attention-augmented LSTM** (56K params) on upper-body pose sequences to classify 5 gesture classes at 84% validation accuracy, deployed via TensorRT on Jetson Orin.
- Engineered a **knowledge distillation** pipeline compressing a **307M-param UniDepth** model into a **390K-param RoI depth head**, using **CKA-based layer analysis** across 24 transformer blocks to select the distillation target, achieving **8.5x faster inference** (51ms → 6ms/frame) and **41% lower BEV MAE** (0.689m → 0.405m).
- Implemented **camera calibration** (checkerboard-based) and manual extrinsic tuning (yaw, pitch, roll) across a 4-camera automotive setup, reducing **projection error by 40%** against tape-measured ground truth.
- Developed multi-camera **Bird's Eye View (BEV)** pipeline using **Inverse Perspective Mapping** and homography, stitching 4 camera feeds into a surround-view with 25% reduction in pixel error over the previous implementation.

Nuvento Systems Pvt. Ltd. **April 2025 - November 2025**
Data Science Intern *Kochi, Kerala*

- Worked on **ExtractIQ**, a document processing platform using **Gemini VLM** and **Facebook BART** to extract, summarize, and classify 200+ PDFs and images via **OCR + NLP** pipelines, reducing manual review time by an estimated 50% (based on analyst feedback).
- Conducted **EDA** across document types (purchase orders, payslips, contracts) to map content variation and class distribution, informing annotation strategy for a 3K-image dataset labeled in **CVAT**.
- Trained **YOLOv11** for document region detection on the custom-annotated dataset; deployed the pipeline on **Microsoft Azure** (App Services) with **Django REST APIs**.

Centre for Airborne Systems, DRDO **November 2023 - December 2023**
Research Intern *Bengaluru, Karnataka*

- Deployed a pretrained **YOLO** object detection model for classification on **NVIDIA Jetson AGX Orin**, optimizing with **TensorRT** FP16 quantization and **DeepStream** to reduce inference latency from 350ms to sub-100ms (3.5x speedup) with minimal accuracy loss on distant objects.

Projects

Bulkhead: Reliable Agent Harness | *Python, Groq API* **GitHub**

- Designed a **tool-calling agent harness** with retry/backoff, timeout enforcement, and loop detection so a broken tool can't hang or silently fail the agent
- Verified reliability with a **benchmark of injected tool failures** (timeouts, malformed output, exceptions), scoring recovery behavior instead of just asserting it works

ShrinkWrap: LLM Cost & Token Optimization Lib | *Python, Anthropic SDK, AWS Bedrock* **GitHub**

- Built a drop-in Python wrapper that intercepts LLM API calls and applies a configurable **reduction pipeline** (prompt caching, context summarization, adaptive `max_tokens`, response caching) with zero semantic degradation

- Defined an **evaluation approach**, a fixed 10-prompt test set benchmarked across Anthropic and **AWS Bedrock**, measuring **26.5% lower token usage and 27.9% lower cost**, with automatic fallback for reliability across backends

ScholarAI: Research Paper Q&A System | *LangChain, FastAPI, ChromaDB, Streamlit* **GitHub**

- Created a **RAG pipeline** with **LangChain** RetrievalQA and **ChromaDB** vector storage over research papers, with dual LLM backends (Gemini + local **LLaMA** via Ollama) and a **FastAPI** + Streamlit interface averaging 500ms query latency

Emulsion: Recipe Similarity Map | *Python, Sentence-Transformers, UMAP* **GitHub**

- Mapped 3,222 recipes into an interactive graph routable by ingredient or technique similarity, combining **TF-IDF**-weighted ingredient overlap with **sentence-transformer embeddings** of cooking instructions
- Reduced blended similarity scores to a 2D layout via **PCA** and **UMAP**, evaluating route quality by ingredient-overlap smoothness and cuisine/course clustering accuracy

Technical Skills

Programming Languages: Python, C, C++, Java, SQL
Deep Learning & CV: PyTorch, TensorFlow, CNNs, LSTMs, Object Detection, Segmentation, Image Processing, OpenCV
Generative AI & NLP: LangChain, RAG, LLMs (Claude, GPT, LLaMA), VLMs, Prompt Engineering & Caching, Vector DBs (ChromaDB), LLM Evaluation, AWS Bedrock
Data Science & ML: scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, Feature Engineering, Statistical Analysis, EDA, Time-Series Forecasting, PySpark
Edge Deployment: NVIDIA Jetson, DeepStream SDK, TensorRT, Model Quantization, Inference Optimization
Tools & Platforms: Git, Linux, Docker, FastAPI, Django, Streamlit, CI/CD, Microsoft Azure, Google Cloud (BigQuery)

Education & Certifications

Vellore Institute of Technology, Chennai **August 2021 - May 2025**
B.Tech. Computer Science and Engineering (Cyber Physical Systems), CGPA: 8.22 *Chennai, India*

Indian Institute of Technology, Chennai **January 2023 - June 2024**
Foundation Level Certificate in Data Science and Programming